



Tuning of PID controllers:

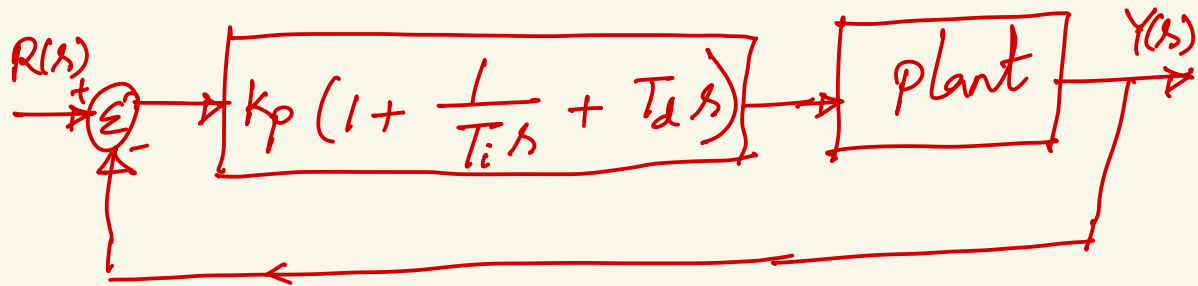
Ziegler - Nichols Tuning :-

An alternative form of the PID controller is as follows:

$$D_{cl}(s) = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

T_i = integral time

T_d = Derivative time



In many situations, the plant

transfer function is not known or too complicated. And it is not apparent what values of K_p , T_i and T_d has to be used.

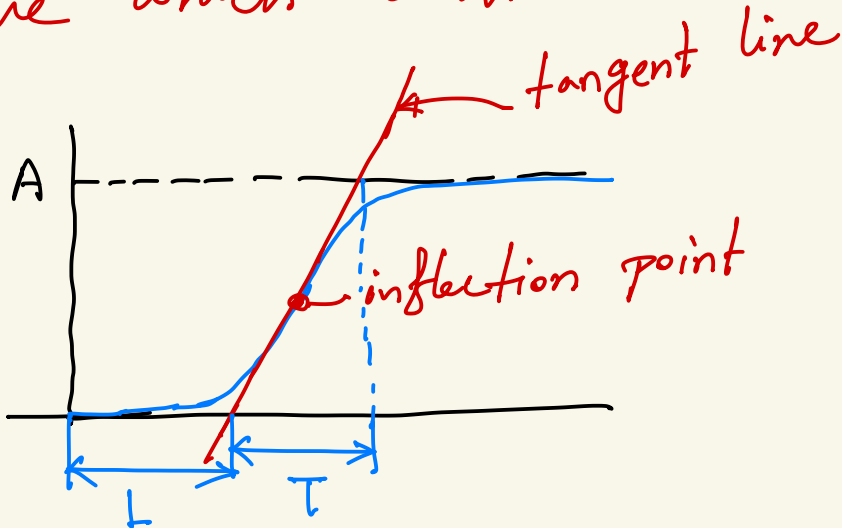
Ziegler and Nichols provided a guideline on what values of K_p , T_i and T_d should be used.

There are generally two methods

- i) Based on a S-shaped process curve (generally a first order system)
- ii) Based on the oscillatory behavior of a system.

Method 1

Ziegler and Nichols noted that the vast majority of chemical or industrial processes can be approximated by the following curve which looks like an 'S'.



L = delay time / Lag time

T = time constant

$$\frac{Y(s)}{U(s)} = \frac{k e^{-Ls}}{T_s + 1} \quad (\text{first order system})$$

The PID gain can be selected based on the L and T values of that specific plant.

	K_p	T_i	T_d
P	$\frac{T}{L}$	∞	0
PI	$0.9 \frac{T}{L}$	$\frac{L}{0.3}$	0
PID	$1.2 \frac{T}{L}$	$2L$	$0.5L$

The PID controller is as follows

$$D_c(s) = k_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$
$$= 1.2 \frac{T}{L} \left(1 + \frac{1}{2Ls} + 0.5Ls \right)$$

$$D_c(s) = 0.6 T \frac{\left(s + \frac{1}{L} \right)^2}{s}$$

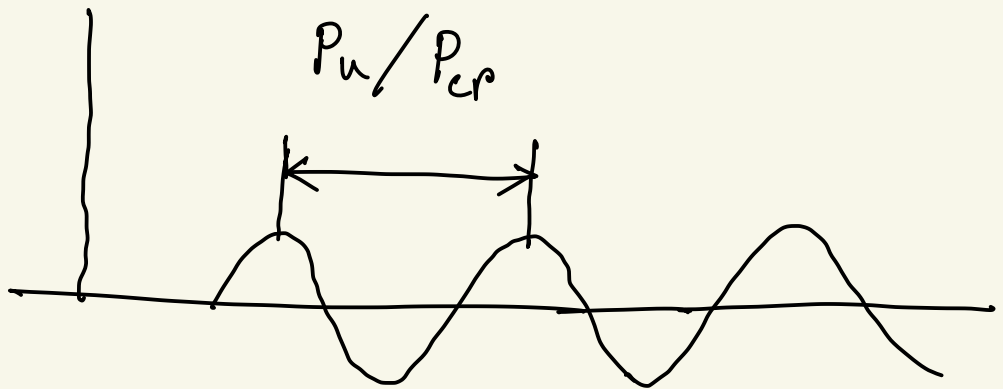
The controller has one pole at the origin and double zeros at

$$s = -\frac{1}{L}$$

Second Method :-

In the second method, the proportional gain k_p is varied from

0 to ∞ until the plant response become totally oscillatory. The specific value of k_p for which the plant response becomes sinusoidal is called the critical gain (k_{cr}) or ultimate gain (k_u).



The period of oscillation is called critical period (P_{cr}) or

ultimate period (P_u). The controller gain is then expressed in terms of these two parameters.

	K_p	T_i	T_d
P	$0.5 K_u$	∞	0
PI	$0.45 K_u$	$\frac{1}{1.2} P_u$	0
PID	$0.6 K_u$	$0.5 P_u$	$0.125 P_u$

$$D_{cl}(s) = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

$$= 0.6 K_u \left(1 + \frac{1}{0.5 P_u s} + 0.125 P_u s \right)$$

$$D_{cl}(s) = 0.075 K_u P_u \frac{\left(s + \frac{4}{P_u} \right)^2}{s}$$

The controller has pole at origin and double zero at $-\frac{4}{P_u}$.